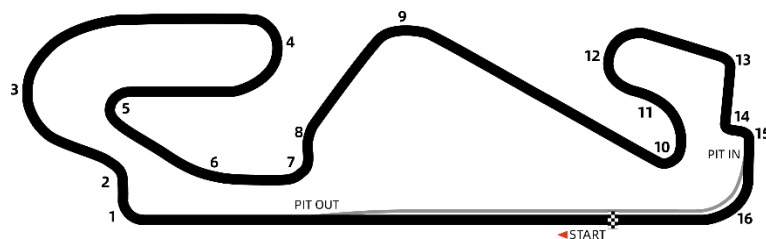


AC REAL-TIME TELEMETRY

CIRCUIT DE BARCELONA-CATALUNYA · PORSCHE 911 RSR · SESSION LOG



// A real-time telemetry pipeline for Assetto Corsa. A custom shared-memory reader captures steering, throttle, brake, RPM and gear data at high frequency and streams it into InfluxDB for visualization in Grafana. This log documents a single session at Circuit de Barcelona-Catalunya, comparing a warm-up lap against a subsequent hotlap in a Porsche 911 RSR (2017). //



Channel Overview



// Three channels recorded in parallel throughout the lap: speed with throttle and brake trace, steering angle, and engine RPM plotted against the active gear. Shift points track cleanly against the RPM ceiling, running from 1st gear at pit exit through 6th on the back straight. //



// Speed – Brake/Throttle. //

Two Laps, One Line

Same corners, two different states of the car - cautious inputs on a cold lap, full commitment once the tires are in.

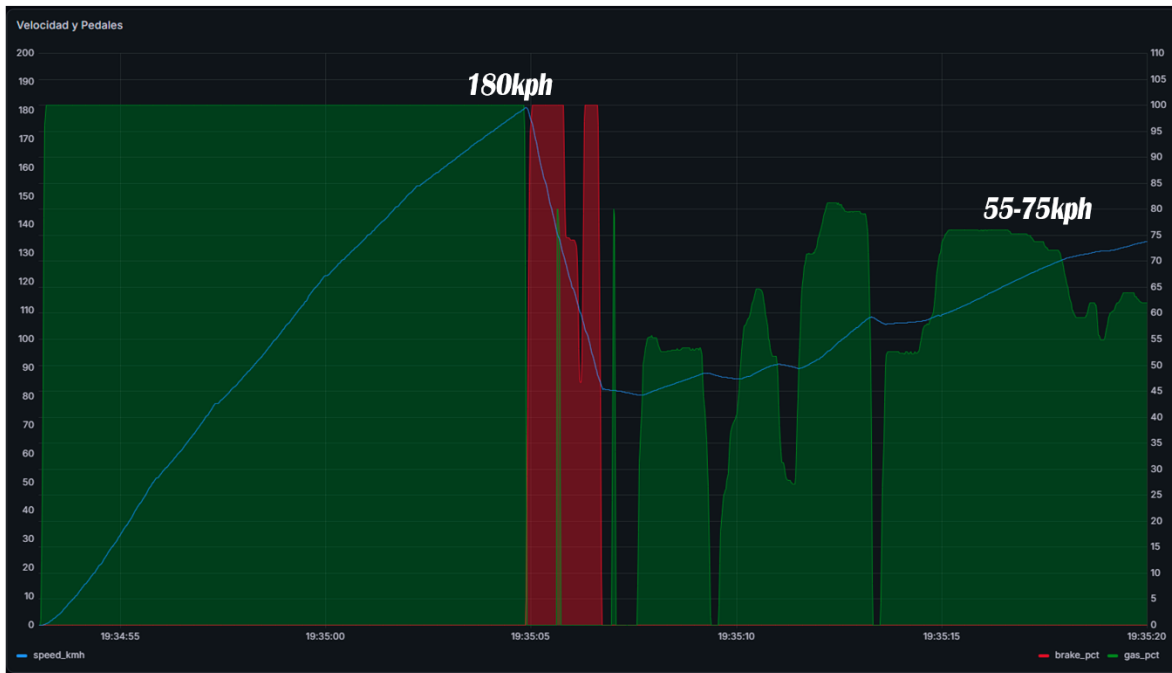


WarmUp Lap

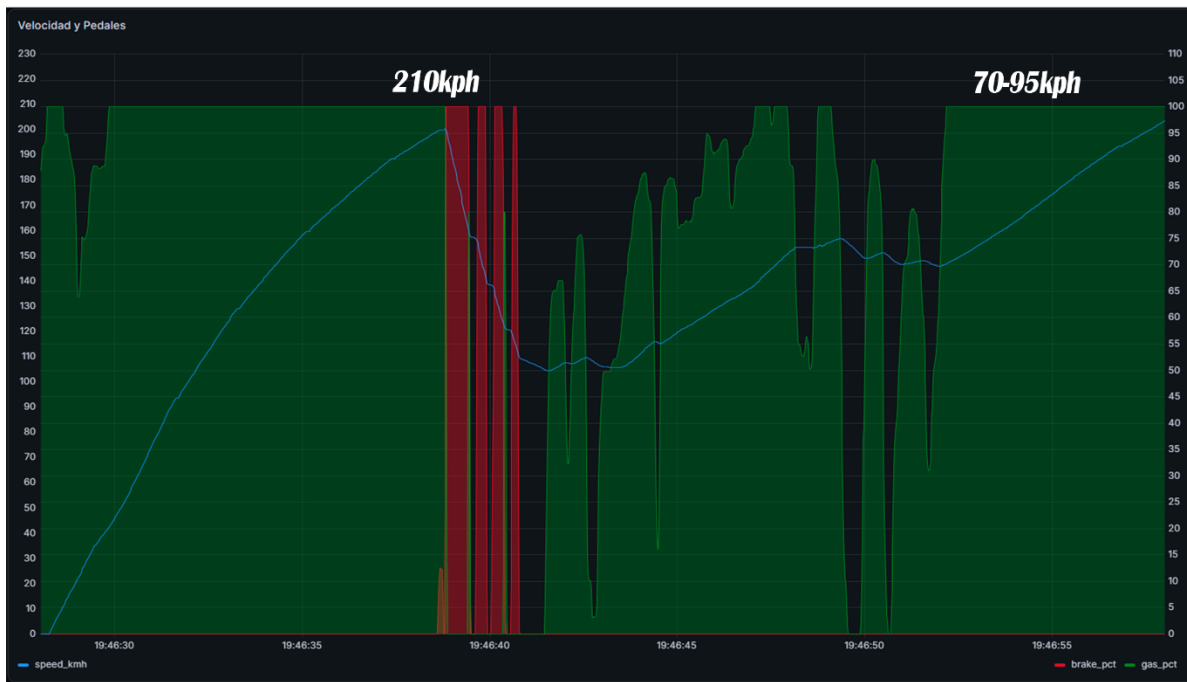


Hotlap

// Steering lock deepens under load. Peak input climbs from 63° on the warm-up lap to 108° at the same T5 on the hotlap, a later, harder turn-in compensating for the extra speed carried into the apex. //



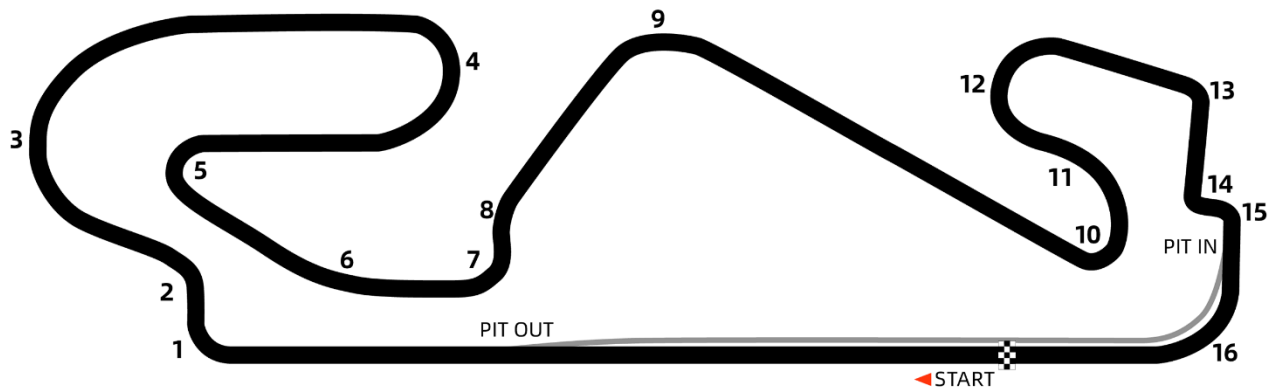
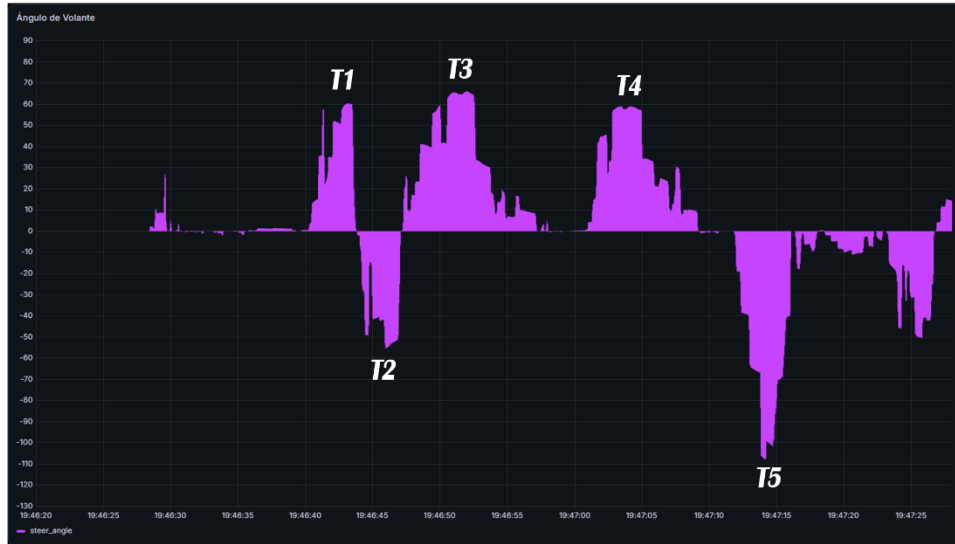
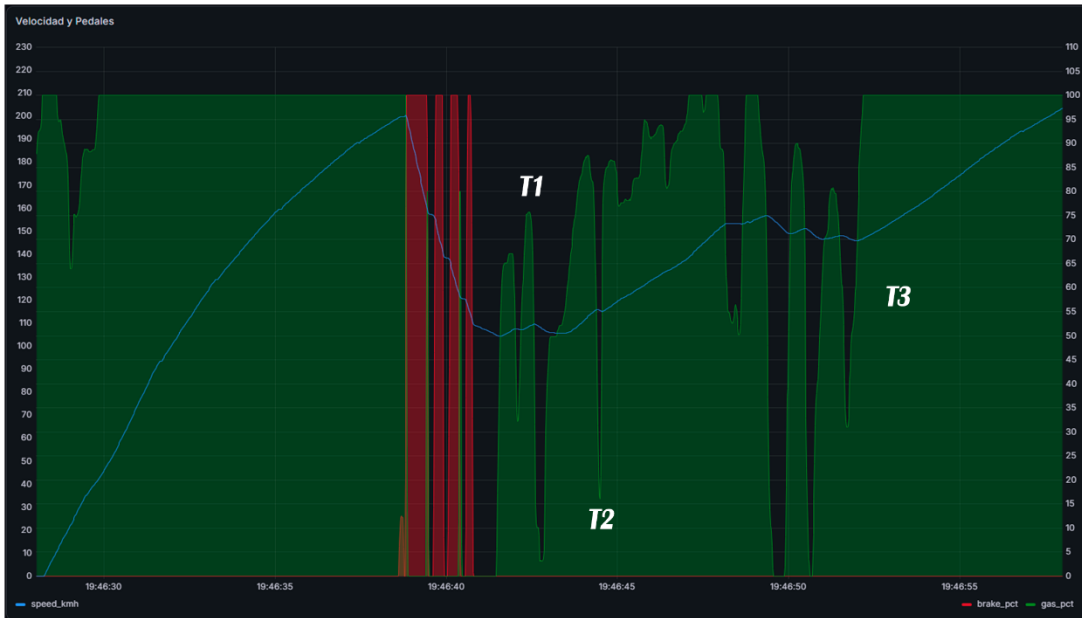
WarmUp Lap



Hotlap

// Top speed before the first braking zone rises from 180 km/h to 210 km/h, and minimum corner speed lifts from the 55–75 km/h band to 70–95 km/h, a faster lap built at both ends of the braking zone, not just on the straight. //

Where It Happens



// Five direction changes - T1 through T5 - define the technical middle sector of the lap. Overlaying the steering and speed traces on the track map pinpoints T5 as the session's sharpest input: the 108° lock from the hotlap, matched to the deepest braking event on the speed trace. The full layout runs 16 numbered corners, with pit entry and exit marked either side of the start/finish line. //

From Sim to Screen

Four stages turn raw simulator physics into the charts on these pages.

Assetto Corsa - Shared memory 

The simulator exposes physics and graphics state through Windows shared memory on every frame.



ac_telemetry.dll - Custom reader

A native module reads the shared memory blocks and extracts the channels used in this report: speed, throttle, brake, steering angle, RPM and gear.



InfluxDB - Time-series store 

Extracted samples are written as time-series points, indexed by session and lap for later queries.



Grafana – Visualization 

Dashboards query InfluxDB directly to render the panels shown throughout this document.



// Building this pipeline end-to-end, from reading raw shared-memory offsets to shaping InfluxDB queries into readable Grafana panels, meant working across the full stack of a real-time data system, not just its surface. It forced a closer look at sampling rates, data types, and how a small ingestion error quietly turns into a misleading chart three steps downstream. Turning steering angle, pedal input, and RPM into a readable story, rather than a wall of numbers, was as much a design problem as a technical one. What started as a way to watch lap data became hands-on practice with time-series storage, dashboarding, and debugging a live pipeline under real, messy conditions. //

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